

VU Shuttle — Campus Transport, Made Visible

A UX case study on designing a real-time shuttle tracker for Vijaybhoomi University students commuting from Karjat.

The Problem

Students commuting between Karjat Railway Station and campus had no way to know where the shuttle actually was, or when it would arrive. Missing the bus, or just not knowing whether to leave the hostel yet, was a recurring source of friction — a theme that, as this project went on, turned out to be shared by nearly every peer working in this space.

The Solution

VU Shuttle — a mobile app centered on four must-have features: **live bus location, daily schedule, nearest-stop finder, and delay/service notifications** — all reachable from a single home screen.

(Insert Screen 3 — Home Page here)

Wireframing → Prototype

Nine core screens were wireframed first — login, home, loading, live map (success and no-data states), schedule, nearest stop, and the return-to-home loop — then built into a working interactive prototype in Figma, and later a fully coded HTML/CSS/JS version to stress-test real interaction.

(Insert wireframe gallery screens here)

Heuristic Evaluation — 3 Fixes Before Real Users Ever Touched It

Before usability testing, I ran a heuristic review and caught three violations:

1. **User Control & Freedom** — the loading screen had no way out if data was slow. *Fix: added a back button.*
2. **Error Recognition & Recovery** — the "No Data Found" screen was a dead end. *Fix: added guidance pointing to the schedule.*
3. **Error Prevention** — a failed login wiped both fields. *Fix: only the password clears now; the mobile number stays filled.*

(Insert before/after heuristic comparison images here)

Usability Testing — 5 Real Users

Each participant was given one task: *"Check the live bus location and arrival time."*

- **4 of 5** completed it easily, no issues.
- **1 participant** hesitated during the loading screen — unsure whether the app was working or frozen, saying *"Wait, is this actually moving or stuck?"* This flagged a **Visibility of System Status** gap: the spinner told users something was happening, but not that it was still working.

Fix identified: a short reassurance line during loading (e.g. "Almost there...") instead of a plain static spinner — a small, low-severity change that closes the loop between what the heuristic review predicted and what a real user actually felt.

Peer Benchmarking — 3 Conversations, One Shared Insight

I met with three students working in the same domain — campus and public transport tracking — plus compared notes with a student in an entirely different domain (Hostel Complaints) for contrast.

The shared root pain point across every conversation: lack of real-time information — the exact same theme that surfaced independently in my own usability testing.

Where approaches differed:

- **Harshal, Kunal, and Yuvraj** (shuttle-tracking peers) built in a **fallback**: if the shuttle is missed, their apps surface private-transport options rather than ending the flow. VU Shuttle currently treats the shuttle as the only path.
- A peer working on **government bus tracking on the same physical route** offered a useful contrast in scale — a fixed, formal public timetable versus a single-route university shuttle.
- Comparing against **Hostel Complaints** (a different domain) was a reminder that information architecture should follow how people describe their problem, not just where it physically happens — prompting me to reconsider whether Nearest Stop should support search by purpose ("stop near library"), not just by name.

What's Next

- Add a lightweight next-step for a missed bus — informational alternate-transport guidance, borrowed from peers' fallback pattern.
- Reconsider purpose-based search in Nearest Stop.
- Ship the loading-screen reassurance cue validated by real usability testing.

Prototype

[Figma prototype link]

